

The logic of arguments - Set 2

TMUA practice questions by topic. Specification Arg1-Arg4. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Consider the statement S : **for all real x , $x^2 - 6x + 10 > 0$.** Which of the following is true?
- A S is false, because $x = 3$ gives $x^2 - 6x + 10 = 0$.
 - B S cannot be decided without knowing x .
 - C S is false, because a quadratic can be negative.
 - D S is true, because $x^2 - 6x + 10 = (x - 3)^2 + 1$.
 - E S is true, because $x^2 - 6x + 10$ has two real roots.
- 2 For real numbers a, b : P : ' $ab = 0$ '; Q : ' $a = 0$ '. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .
- 3 Consider the statement: **if a and b are irrational, then $a + b$ is irrational.** Which of the following pairs is a counterexample?
- A $a = 1, b = -1$
 - B $a = \sqrt{2}, b = -\sqrt{2}$
 - C $a = \pi, b = \sqrt{2}$
 - D $a = \sqrt{2}, b = 2\sqrt{2}$
 - E $a = \sqrt{2}, b = \sqrt{3}$
- 4 Let S be the statement: **every even number greater than 2 can be written as the sum of two primes.** A single counterexample to S would be
- A an odd number that is not the sum of two primes
 - B an even number greater than 2 that can be written as the sum of two primes in two different ways
 - C two primes whose sum is odd
 - D an even number that is the sum of two primes
 - E an even number greater than 2 that is not the sum of any two primes

- 5 For a real number x , P : ' $x > 0$ '. Q : ' $x^3 > 0$ '. Which one of the following is true?
- A P is necessary and sufficient for Q.
 - B P is necessary but not sufficient for Q.
 - C P is sufficient but not necessary for Q.
 - D P is neither necessary nor sufficient for Q.
- 6 Which of the following statements is/are true?
- I For every real number x , there exists a real number y such that $y > x$.
 - II There exists a real number y such that for every real number x , $y > x$.
 - III For every real number x , there exists a real number y such that $xy = 1$.
- A none of them
 - B I only
 - C II only
 - D III only
 - E I and II only
 - F I and III only
 - G II and III only
 - H I, II and III
- 7 A teacher says: '**You will pass only if you revise.**' Which of the following statements means the same thing?
- A If you pass, then you revised.
 - B If you revise, then you will pass.
 - C You will pass if you revise.
 - D If you do not pass, then you did not revise.
 - E You will pass if and only if you revise.

- 8** Consider the statement: ****there exists a real number k such that for all real x , $x^2 + kx + 1 > 0$.**** Which of the following is true?
- A** The statement is true; for example $k = 0$ works.
 - B** The statement is false because $x^2 + kx + 1$ can be negative for large x .
 - C** The statement is true; every value of k works.
 - D** The statement is false because for every k there is an x with $x^2 + kx + 1 \leq 0$.
 - E** The statement is true; for example $k = 2$ works.
- 9** Which of the following statements is/are true about real numbers?
- I** For all x and all y , if $x < y$ then $x^2 < y^2$.
 - II** For all x there exists y such that $x < y$ and $x^2 < y^2$.
 - III** There exist x and y with $x < y$ and $x^2 > y^2$.
- A** none of them
 - B** I only
 - C** II only
 - D** III only
 - E** I and II only
 - F** I and III only
 - G** II and III only
 - H** I, II and III
- 10** A statement reads: ****for every positive integer n , there is a prime p with $n < p < 2n$.**** Which of the following is its negation?
- A** For every positive integer n , there is a prime p with $p \leq n$ or $p \geq 2n$.
 - B** There is a positive integer n such that no prime p satisfies $n < p < 2n$.
 - C** There is a positive integer n and a prime p with $p \leq n$ or $p \geq 2n$.
 - D** There is a prime p such that for every positive integer n , $n < p < 2n$ fails.
 - E** For every positive integer n , no prime p satisfies $n < p < 2n$.

11 Which of the following statements is/are logically equivalent to '**if it is raining, then the ground is wet**'?

- I** It is not raining, or the ground is wet.
- II** If the ground is not wet, then it is not raining.
- III** If the ground is wet, then it is raining.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

12 Consider the statement S : ****for all real x , if $x > 0$ then $x + \frac{1}{x} \geq 2$.**** Which of the following statements is/are true?

- I** S is true.
- II** The converse of S (for all real x , if $x + \frac{1}{x} \geq 2$ then $x > 0$) is true.
- III** The contrapositive of S is true.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III